AMITY SCHOOL OF ENGINEERING AND TECHNOLOGY

AMITY UNIVERSITY

----------BENGALURU----------

<u>SOUECE CODE MANAGEMENT</u>

CSE2015

SUBMITTED BY:

NAME: SPOORTHI V SHARMA

SEN: A866175124105

COURSE: BTECH CSE(AIML)

SLOT: L25+26

SUBMITTED TO:

MS MANISHA NAIR.

1.INTRODUCTION:

1.What is Git?

- **Define Git as a version control system**
- **Explain its purpose (tracking changes, collaboration, etc.)**

❖ **Git** is a version control system used to track changes in files and manage source code during software development.

- It helps in tracking changes made to files.
- It allows multiple users to work on the same project
- It maintains different versions of the project.

2.What is Version Control?

- **Explain the concept of version control**
- **Importance in software development**

❖ **Version control** is a system that records changes to files over time.

It allows you to:

- Go back to previous versions
- Compare changes
- Work safely without losing data

❖ **Importance in software development:**

- Helps multiple developers work together
- Prevents overwriting each other's work
- Makes debugging easier
- Keeps a history of all change

3.What is GitHub?

- **Define GitHub as a platform for hosting repositories**
- **Mention its role in collaboration**

❖ GitHub is an online platform used to host Git repositories.

❖ It allows developers to:

- Store their code online
- Share projects
- Collaborate with others

- It provides features like pull requests, issues, and project management tools.

4.Difference Between Git and GitHub

- **Write atleast 3-5 clear differences**

Git	GitHub
1. Git is a software/tool	2. GitHub is a cloud platform/service
3. Works locally on your computer	4. Works online on the internet
5. Used for tracking changes	6. Used for sharing and collaboration
7. Does not require internet	8. Requires internet
9. Developed by Linus Torvalds	10.Owned by Microsoft

2. BASIC GIT COMMANDS:

- Include a list of all Git/Git Bash & vi Editor commands performed during the previous lab.
- You may refer to the commands covered in class and the provided material.
- For every command, write an explanation of what it does.
- After each command and its description, attach a relevant screenshot showing:
- Command execution in terminal (Git Bash)

1. GIT CONFIGURATION COMMANDS:

- **git --version:** checks the installed git version on the system.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git --version
git version 2.50.0.windows.1
```

- **git config --global user.name "Your Name":**Sets the global username for git commits.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git config --global user.name "SPOORTHI"
```

- **git config --global user.email email@gmail.com:** Sets the global email for Git commits.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git config --global user.email "sharmaspoorthi02@gmail.com"
```

- **git config user.name:** Displays the configured Git username.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git config user.name
SPOORTHI
```

- **git config user.email :** Displays the configured Git email.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git config user.email
sharmaspoorthi02@gmail.com
```

- **git config --list :** Shows all Git configuration settings.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=SPOORTHI
user.email=sharmaspoorthi02@gmail.com
user.name=SPOORTHI
core.editor=vi
```

- **git help** : Displays help/manual for Git commands.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git help
usage: git [-v | --version] [-h | --help] [-C <path>] [-c <name>=<value>]
           [--exec-path[=<path>]] [--html-path] [--man-path] [--info-path]
           [-p | --paginate | -P | --no-pager] [--no-replace-objects] [--no-lazy
-fetch]
           [--no-optional-locks] [--no-advice] [--bare] [--git-dir=<path>]
           [--work-tree=<path>] [--namespace=<name>] [--config-env=<name>=<envva
r>]
           <command> [<args>]

These are common Git commands used in various situations:

start a working area (see also: git help tutorial)
   clone      Clone a repository into a new directory
   init       Create an empty Git repository or reinitialize an existing one

work on the current change (see also: git help everyday)
   add        Add file contents to the index
   mv         Move or rename a file, a directory, or a symlink
   restore    Restore working tree files
   rm         Remove files from the working tree and from the index

examine the history and state (see also: git help revisions)
   bisect     Use binary search to find the commit that introduced a bug
   diff       Show changes between commits, commit and working tree, etc
   grep       Print lines matching a pattern
   log        Show commit logs
   show       Show various types of objects
   status     Show the working tree status

grow, mark and tweak your common history
   backfill   Download missing objects in a partial clone
   branch     List, create, or delete branches
   commit     Record changes to the repository
   merge      Join two or more development histories together
   rebase     Reapply commits on top of another base tip
   reset      Reset current HEAD to the specified state
   switch     Switch branches
   tag        Create, list, delete or verify a tag object signed with GPG

collaborate (see also: git help workflows)
   fetch      Download objects and refs from another repository
   pull       Fetch from and integrate with another repository or a local branch
   push       Update remote refs along with associated objects

'git help -a' and 'git help -g' list available subcommands and some
concept guides. See 'git help <command>' or 'git help <concept>'
to read about a specific subcommand or concept.
See 'git help git' for an overview of the system.
```

2. Navigation Commands

- **pwd :** Shows the current working directory (present folder location).

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ pwd
/c/Users/SPOORTHI
```

- **cd folder_name/location**: Changes the current directory to another folder.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cd D:

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
```

3.File and Folder Listing Commands

- **ls :** Shows the current working directory (present folder location).

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ ls
'$RECYCLE.BIN'/    SCM_3                        file3.txt          pull/
 4.txt             SPOO1234/                    folder1/           pullrequest/
 ASSIGN2           SVS12/                       folder2/           request/
 ASSIGN2.txt      'System Volume Information'/  folder3/           s122/
 CA-1/             V/                           fork/              s122.text
 CAL/              abcd12/                      html/              summer_4aug/
 E123/             cd/                          main_flolder_SCM/  temperature/
 EVALUATION_2/     cv.txt                       nano.2088.save     todolist/
 Project_Team12/   file                         nature/
 RR1234/           file1.txt                    practice123/
 RR678/            file2.txt                    prctice123/
```

- **ls -l :** Changes the current directory to another folder.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ ls -l
total 14
drwxr-xr-x 1 SPOORTHI 197121   0 Jun 27  2025 '$RECYCLE.BIN'/
-rw-r--r-- 1 SPOORTHI 197121   0 Mar 17 13:38  4.txt
-rw-r--r-- 1 SPOORTHI 197121  38 Mar 17 14:12  ASSIGN2
-rw-r--r-- 1 SPOORTHI 197121  42 Mar 17 14:18  ASSIGN2.txt
drwxr-xr-x 1 SPOORTHI 197121   0 Oct 23 15:45  CA-1/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul  7  2025  CAL/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul  7  2025  E123/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 21  2025  EVALUATION_2/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 28  2025  Project_Team12/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 14  2025  RR1234/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul  7  2025  RR678/
-rw-r--r-- 1 SPOORTHI 197121  38 Mar 17 14:16  SCM_3
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 28  2025  SPOO1234/
drwxr-xr-x 1 SPOORTHI 197121   0 Jun 30  2025  SVS12/
drwxr-xr-x 1 SPOORTHI 197121   0 Jun 20  2025 'System Volume Information'/
drwxr-xr-x 1 SPOORTHI 197121   0 Aug  2  2025  V/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 28  2025  abcd12/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 14  2025  cd/
-rw-r--r-- 1 SPOORTHI 197121 124 Aug  2  2025  cv.txt
-rw-r--r-- 1 SPOORTHI 197121   0 Mar 17 13:38  file
-rw-r--r-- 1 SPOORTHI 197121   0 Mar 17 13:59  file1.txt
-rw-r--r-- 1 SPOORTHI 197121   0 Mar 17 13:38  file2.txt
-rw-r--r-- 1 SPOORTHI 197121   0 Mar 17 13:38  file3.txt
drwxr-xr-x 1 SPOORTHI 197121   0 Mar 17 13:45  folder1/
drwxr-xr-x 1 SPOORTHI 197121   0 Mar 17 13:45  folder2/
drwxr-xr-x 1 SPOORTHI 197121   0 Mar 17 13:45  folder3/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 28  2025  fork/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 20  2025  html/
drwxr-xr-x 1 SPOORTHI 197121   0 Mar 17 13:49  main_flolder_SCM/
-rw-r--r-- 1 SPOORTHI 197121 151 Mar 23 11:04  nano.2088.save
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 30  2025  nature/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 14  2025  practice123/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 14  2025  prctice123/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 28  2025  pull/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 28  2025  pullrequest/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 28  2025  request/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul  7  2025  s122/
-rw-r--r-- 1 SPOORTHI 197121  67 Jul  7  2025  s122.text
drwxr-xr-x 1 SPOORTHI 197121   0 Aug  4  2025  summer_4aug/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 28  2025  temperature/
drwxr-xr-x 1 SPOORTHI 197121   0 Jul 20  2025  todolist/
```

- **ls -a :** Lists all files including hidden files (files starting with .)

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ ls -a
'$RECYCLE.BIN'/   E123/                        V/          folder2/            pull/
 ./               EVALUATION_2/               abcd12/     folder3/            pullrequest/
 ../              Project_Team12/             cd/         fork/               request/
 .git/            RR1234/                     cv.txt      html/               s122/
 4.txt            RR678/                      file        main_flolder_SCM/   s122.text
 ASSIGN2          SCM_3                       file1.txt   nano.2088.save      summer_4aug/
 ASSIGN2.txt      SPOO1234/                   file2.txt   nature/             temperature/
 CA-1/            SVS12/                      file3.txt   practice123/        todolist/
 CAL/            'System Volume Information'/ folder1/   prctice123/
```

- **ls -d */:** Lists only directories (folders) in the current location.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ ls -d*/
ls: unknown option -- *
Try 'ls --help' for more information.

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ ls -d */
'$RECYCLE.BIN/'/    SPOO1234//                  folder3//           pullrequest//
 CA-1//             SVS12//                     fork//              request//
 CAL//             'System Volume Information/'/ html//            s122//
 E123//             V//                         main_flolder_SCM//  summer_4aug//
 EVALUATION_2//     abcd12//                    nature//            temperature//
 Project_Team12//   cd//                        practice123//       todolist//
 RR1234//           folder1//                   prctice123//
 RR678//            folder2//                   pull//
```

4. File Operations:

- **touch filename:** Creates a new empty file.
- **touch file1 file2 file3:** Creates multiple files at once.
- **rm filename:** Deletes a file.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ touch scm_labfile

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ touch file1 file2 file3

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ rm scm_labfile
```

5. Folder Operations:

- **mkdir folder_name:** Creates a new folder.
- **mkdir folder1 folder2 folder3:** Creates multiple folders at once
- **rmdir folder_name:** Deletes an empty folder.
- **rm -r foldername:** Deletes a folder and its contents (non-empty folder).

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ mkdir main_flolder_SCM

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ cd main_folder_SCM
bash: cd: main_folder_SCM: No such file or directory

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ mkdir main_folder_SCM

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ cd main_folder_SCM

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d/main_folder_SCM
$ touch abc.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d/main_folder_SCM
$ touch 123.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d/main_folder_SCM
$ mkdir sub1 sub2

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d/main_folder_SCM
$ cd..
bash: cd..: command not found

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d/main_folder_SCM
$ cd ..

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ rmdir main_folder_SCM
rmdir: failed to remove 'main_folder_SCM': Directory not empty
```

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 /d
$ rm -r main_folder_SCM
```

6.viewing file content:

- **cat filename:** display the contents of a file in the terminal

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat  > filename
```

7. write text to a file using echo

- **echo "text-content" > filename:** writes text to a file. If the file does not exist ,it will be created
- **echo "text" >> file name:** appends text to an existing file without deleting previous content.

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ echo "THIS IS THE TASK 1 DATE-24/03/2026"> file1.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ echo "learning vi editior">>file1.txt
```

8. Writing text using cat:

- **cat > file name:** creates a file and allows us to type taxt directly in the terminal
- **cat >> filename:** adds text to an existing file without deleting previous content

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat file1.txt
THIS IS THE TASK 1 DATE-24/03/2026
learning vi editior

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat >file2.txt
THIS IS FILE 2
LEARNING VI EDITOR
I AM LEARNING GIT AND VI
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat file2.txt
THIS IS FILE 2
LEARNING VI EDITOR
I AM LEARNING GIT AND VI
```

9. Copy content fron one file to another:

- **cat file1.txt > file2.txt:** copies the contant of file1.txt into file2.txt

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat file1.txt > file2.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat file2.txt
THIS IS THE TASK 1 DATE-24/03/2026
learning vi editior
```

10.Editing a file using nano editor:

- **nano file.txt:** opens the nano text editor to create or edit a file**.**

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ nano file.txt
```

11.Vi+ git configuration:

- **git var GIT _EDITOR:** check the default editor via git
- **git config –global core.editor "vi":** set vi editor as the default editor for git bash

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git config --global core.editor "vi"

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git var GIT_EDITOR
vi

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ git config --global user.name "SPOORTHI" git

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ config --global user.email "sharmaspoorthi02@gmail.com"
bash: config: command not found

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ config --global user.email "sharmaspoorthi02@gmail.com"
bash: config: command not found

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$  git config --global user.email "sharmaspoorthi02@gmail.com"
```

- **vi filename.txt: opens a file in vi editor**

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ vi SPOORTHI.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat SPOORTHI.txt
NAME : SPOORTHI V SHARMA
COURSE : BTECH CSE(AIML)
I AM INTRESTED IN PROGRAMMING
I AM LEARNING LINUX AND GIT
I STUDY AT AMITY UNIVERSITY BENGALURU
```

12.insert +navigation commands in vi editor:

- **i:** inserts text before the current cursor
- **I:** inserts text at the beginning of the line
- **a:** insert text aster the current cursor
- **A:** insert text at the end of the current line
- **o :** creates a new line for text entry below cursor location and switches to insert mode.
- **O**: creates a new line for text entry above cursor location and switches to insert mode
- Press **esc** to return to normal mode.
- **:w-** save files
- **:q-** quit
- **:wq** – save and quit
- **:x** – save and quit
- **:q!** – quit without saving

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ vi SPOORTHI.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat SPOORTHI.txt
NAME : SPOORTHI V SHARMA
COURSE : BTECH CSE(AIML)
I AM INTRESTED IN PROGRAMMING
I AM LEARNING LINUX AND GIT
I STUDY AT AMITY UNIVERSITY BENGALURU

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ vi SPOORTHI.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat SPOORTHI.txt
THIS IS THE TOP LINE
NAME : SPOORTHI V SHARMA
COURSE : BTECH CSE(AIML)
I AM INTRESTED IN PROGRAMMING
I AM LEARNING LINUX AND GIT
I STUDY AT AMITY UNIVERSITY BANGLORE -EDITED LINE
THIS IS THE BOTTOM LINE
```

13. DELETE commands in vi editor:

- **x :** delete a character

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ vi SPOORTHI.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat SPOORTHI.txt
THIS IS THE TOP LINE

NAME : SPOORTHI V SHARMA
COURSE : BTECH CSE(AIML)
I AM INTRESTED IN PROGRAMMING
I AM LEARNING LINUX AND G
I STUDY AT AMITY UNIVERSITY BANGLORE -EDITED LINE
THIS IS THE BOTTOM LINE
```

- **dd :** delete current line

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ vi SPOORTHI.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat SPOORTHI.txt
THIS IS THE TOP LINE

NAME : SPOORTHI V SHARMA
COURSE : BTECH CSE(AIML)
I AM INTRESTED IN PROGRAMMING
I STUDY AT AMITY UNIVERSITY BANGLORE -EDITED LINE
THIS IS THE BOTTOM LINE
```

d$: delete from cursor to end of line

```
SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ vi SPOORTHI.txt

SPOORTHI@LAPTOP-MG8SDVJP MINGW64 ~
$ cat SPOORTHI.txt
THIS IS THE TOP LINE

NAME : SPOORTHI V SHARMA
COURSE : BTECH CSE(AIML)
I AM INTRESTED IN PROGRAMMING
```

- **dθ:** delete from cursor to start of line